

Figure 7: Multi threaded application with one boss and 4 worker threads

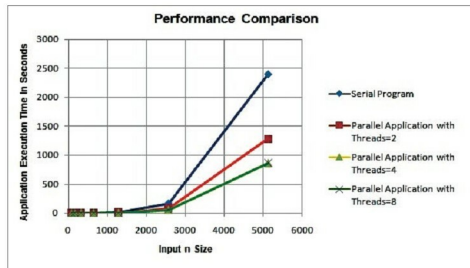


Figure 8: Performance comparison

```
}
*/
}
```

The execution is as follows:

```
$ cc -o matrix_pthr_5120_2 matrix_mul_pthreads.c -pthread
$ time ./matrix_pthr_5120_2
```

This will create a multi-threaded application with two worker threads and one boss thread. The performance can be monitored by `htop` as shown in Figure 5. The two worker threads are running in parallel, simultaneously, on

Cores 2 and 4, while the main thread (boss) is in a waiting state as shown in Figure 5. The total CPU utilisation is 55.4 per cent. The application performance is better than the serial application. If you replace the statement `#define nthreads 2` in the above program with `#define nthreads 4` and recompile it, then it generates a multi-threaded application with four worker threads and one boss thread as shown in Figures 6 and 7. The four worker threads are running simultaneously on Cores 1, 2, 3, and 4 respectively, in parallel. It is very clear the CPU utilisation has improved to more than 99 per cent. So the application's performance is very good.

We have executed the serial application and multi-threaded applications by taking $n = 80, 160, 320, 640, 1280, 2560, 5120$ and $nthreads = 2, 4, 8$ in the case of multi-threaded applications. A comparison of the results is shown in Figure 8. In terms of execution time, the multi-threaded application clearly outperforms the serial application's performance. So, with the Pthreads API we can develop

parallel applications to get a higher performance from the multi-core microprocessor. **END**

References

- [1] 'Advanced Linux Programming', a book by Mitchell, Jeffrey Oldham and Alex Samuel
- [2] <https://computing.lln.gov/tutorials/pthreads/>

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